Evaluating and Extending an RNN-based POS Tagger

- **Goal:** compare models and evaluate extensions to an RNN POS tagging model.
- **Data:** Universal Dependencies datasets.
- **Languages:** English, Spanish, Russian, Swedish, Basque, Chinese, Finnish, Turkish, Japanese, Latin.
- **Baseline:** Majority tag baseline used as a reference. 16–30% accuracy

Model Design

- **Model architecture:**

GRU vs LSTM

Unidirectional vs Bidirectional RNN

- **Training:**

Dropout regularization [0.0, 0.2, 0.5]

Random token masking during training

- **Tagging:**

UPOS vs XPOS tagsets

- **Additional experiment:**

Multilingual training (English + Spanish)

Results

- **Neural models**

GRU / LSTM models: 80–95% accuracy

- **Other experiments**

Dropout and masking show no major performance change

UPOS and XPOS produce similar results.

- **Observations**

Bidirectional models generally **improve performance**

GRU and LSTM show very **similar** results

Highest accuracies observed in Japanese and Russian (94%)

Lower performance in Turkish, Basque and Finnish

Interpretation

- **Model architecture:**

Bidirectional models improve performance because they use both left and right context when predicting POS tags.

- **Language structure:**

Languages such as Turkish, Basque and Finnish are more morphologically complex (agglutinative), increasing word variability and making tagging more difficult.

- **Dataset size:**

UD datasets are not the same size for all languages.
Languages with more training data usually give better and more stable results, while smaller datasets can lead to more variation in performance.

Additional Experiment

```
Word: want
Closest words:
want
promised
buscando
conflict
lashed
excreción
focusing
dijo
helping
distant
```

```
Word: estar
Closest words:
estar
Usera
detonated
aprobar
existed
shark
Barno
asaltado
pintar
Zhougu
```

```
Word: pero
Closest words:
pero
servidor
but
hienas
dudaba
extras
Cossío
salvador
palmesana
onesie
```